DAT158 – Machine learning and advanced algorithms – Høgskulen på Vestlandet

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Exercise number 5

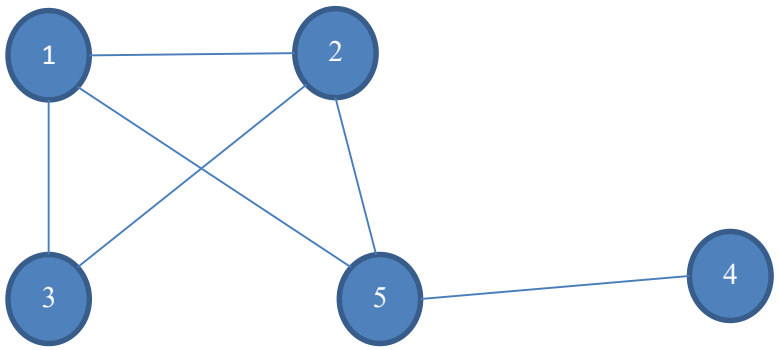
Problem 1 (Part of compulsory)

The graph below is 3-colorable. Use the algorithm that colors a 3-colorable graph with $O(\sqrt{n})$ colors, Chapter 6 to color the graph. Every time you have a choice between nodes, choose the lowered number one. Notice, it is important to understand the algorithm, not find the best coloring.

	1	2	3	4	5	6	7	8	9
1			1	1		1		1	
2							1		1
3	1			1		1			
4	1		1						
5						1			1
6	1		1		1			1	
7		1						1	1
8	1					1	1		
9		1			1		1		

Problem 2 (Part of compulsory)

A vertex cover of a graph is a subset of the vertices, $C \subseteq V$ such that every edge has at least one endpoint in C, that is $\forall(i.j) \in E$, either $i \in C$ or $j \in C$. We want to find a minimum vertex cover.



a) Formulate the problem above as an integer linear program (ILP).

b) Formulate the LP relaxation of the ILP above.

c) Formulate the Dual version of the LP problem above.

d) We have a large graph with 10.000 vertices. We solve the LP-problem and get the solution 500 (number of vertices in the vertex cover). What can we say about the solution of the original problem in a)?